

Urologic Complications of Renal Transplantation

- Renal transplantation is the **most cost-effective treatment for end-stage renal disease (ESRD)**, improving survival and quality of life.
- Urologic complications are the **commonest surgical adverse event after transplantation**, at 8% to 9.5%, with high morbidity and cost.
- The **majority arise from the transplanted ureter and the ureterovesical anastomosis** — the fact that organises the whole subject.

HAEMATURIA

- Immediate postoperative haematuria comes **almost always from the ureterovesical anastomosis**, resolves over several days, and rarely needs bladder irrigation.
- It is **commoner after the Leadbetter anastomosis** than after other types.
- **Persistent haematuria after allograft biopsy means an arteriovenous-collecting system fistula** until proven otherwise: duplex ultrasound diagnoses, magnetic resonance angiography confirms.
- Most such fistulae **close spontaneously**; those that do not need **highly selective embolisation**.
- Asymptomatic microscopic haematuria follows American Urological Association guidance, with **one transplant-specific addition: BK virus** — suspected from viral cytopathic changes on cytology, confirmed by urinary BK titres.

URETERAL STENT MANAGEMENT

Timing of Stent Removal

- Randomised trials settled that **routine stenting across the ureterovesical anastomosis reduces urine leak and ureteral stricture**.
- The cost is **bacterial colonisation** — roughly a quarter of stents, regardless of dwell time. The resulting urinary tract infection (UTI) threatens graft infection and graft failure, which is why timing matters at all.
- Meta-analysis shows **removal before the third postoperative week reduces UTI without increasing leak or stenosis**; repeat meta-analysis in 2022 confirmed it and fixed the optimum at **14 to 21 days**.
- The counterweight: **removal before 14 days increases urinary leak**. Hence a window, not a date.
- Timing remains **individualised** — surgeon satisfaction with the anastomosis, and recipient risk factors: diabetes mellitus, transplant revision, current UTI.
- Prophylaxis at removal is **unresolved**. Patients already on trimethoprim-sulfamethoxazole for *Pneumocystis jirovecii* pneumonia appear protected even where resistance is high, and **additional prophylaxis adds nothing**; those not on it may benefit from cover at removal.

Retained Stent

- Two mechanisms: **the ureteroneocystostomy suture has caught the stent**, or **encrustation** (incidence up to 5.7%, a function of prolonged dwell time).
- A trapped suture leaves two options: **wait for suture dissolution, or transect the stitch endoscopically**.
- **Allograft denervation abolishes the classic obstructive symptoms**. The encrusted stent therefore presents as **progressive decline in renal function, falling urine output, recurrent UTIs**; rarely a vague ache over the allograft from peritoneal irritation.

Prevention

- Stent encrustation is the **textbook illustration of Reason's Swiss cheese model** — a chain of latent errors: undocumented insertion, poor team communication, and inadequate counselling on the stent's temporary function, its complications, and the need for timely removal.
- Log books, stent cards, computerised and web-based registries and smartphone applications have all been studied; **none is completely reliable**.

Diagnosis and Treatment

- **Computed tomography (CT) in every case**, to define encrustation burden and distribution and thereby the approach.
- First step is **urinary diversion by percutaneous nephrostomy** — decompression, recovery of function, and antegrade access in one move.
- Clearance then uses **percutaneous nephrolithotomy, retrograde intrarenal surgery and extracorporeal shock wave lithotripsy in combination**.

KEY POINTS: HAEMATURIA AND URETERAL STENT MANAGEMENT

- Most postoperative haematuria is self-limiting.
- Consider duplex ultrasound for severe or persistent haematuria after renal biopsy, to exclude arteriovenous fistula.
- A ureteral stent decreases ureteral stricture and urinary leak.
- Individualised stent removal around 3 weeks after transplantation.

URINARY LEAK

- **The commonest early urologic complication**, at 1.2% to 8.9%.
- The dominant mechanism is **compromise of the distal ureteric blood supply during procurement**, causing ureteric necrosis. Risk rises with advanced donor age, extended-criteria donors, multiple allograft arteries, and **failure to reimplant the lower polar artery**.

BOX 92.1 ADDITIONAL CAUSES OF URINARY LEAK

- An excessively long ureter
- Premature removal of bladder and/or ureteric drainage
- Technical problem — suture dehiscence, ureteric twisting, kinking
- Acute urinary retention or bladder catheter obstruction
- Necrosis of renal parenchyma
- Parenchymal perforation during double-J stent placement

- The box divides into **three mechanisms** — **ischaemia, technical error, and back-pressure from failed drainage**. Each is preventable, which is why leak rates are surgeon-dependent.

Diagnosis

- With a drain: **high drain output, falling urine output, rising serum creatinine**.
- Without one: **wound leakage, wound dehiscence, scrotal swelling, pelvic or abdominal pain**.
- Ultrasound or non-enhanced CT finds the collection; **a fluid creatinine far above the serum creatinine** — on aspirate, drain fluid or wound discharge — proves it is urine.

Treatment

- The principle is to **divert urine away from the extravasation**, protect graft function and stabilise the patient: Foley catheter plus percutaneous nephrostomy.
- The nephrostomy is **both diagnostic (antegrade nephrostogram) and therapeutic (antegrade stent access)** where a stent is absent or non-functional.
- **Retrograde stenting is the harder route** — angulation to the neo-orifice, orifice oedema, and the risk of further disrupting the anastomosis.
- Percutaneous drainage for a **substantial urinoma, obstruction caused by it, or infection**.
- Small anastomotic leaks: **contrast studies every 1 to 2 weeks**, a Foley catheter to aid healing, and removal of catheter and nephrostomy once resolved. **Follow diligently — ureteric stenosis follows later.**
- **Explore early for a proximal large leak, or a leak persisting despite maximal drainage.** Four reasons: conservative success is low, adhesions have not formed, fibrosis is minimal, and later stenosis becomes less likely.
- Reconstruction: **tension-free ureteroneocystostomy if the remnant is well vascularised**; otherwise ureteroureterostomy to the ipsilateral native ureter, a bladder flap, or pyelovesicostomy.

URETERAL STRICTURE

- **The predominant cause of ureteral obstruction beyond 3 months**, affecting 0.5% to 9%.
- **The primary cause is ischaemia** — which is why every risk factor below is a perfusion insult or a healing insult.
- Risk factors: **cold ischaemia above 435 min**, acute rejection, postoperative urinoma or abscess, delayed graft function, multiple allograft arteries, advanced donor age, and — unique to transplantation — **BK virus and cytomegalovirus**.
- **Stricture rates do not differ by anastomotic technique**, nor between brain-dead and donation-after-cardiac-death donors.

TABLE 92.1 COMMON CAUSES OF URETERAL OBSTRUCTION

Early obstruction (<3 months)	Late obstruction (>3 months)
Technical error at ureteroneocystostomy	Stones
Forgoing a ureteral stent	Ureteral strictures
Anastomotic oedema	Lymphocele
Redundant ureter	Fibrosis related to immunosuppressant medications
Extrinsic compression — lymphocele, haematoma, abscess	

- The table is a **timeline, not a list**: early causes are technical and reversible, late causes are ischaemic, metabolic or drug-related.
- **Lymphocele appears in both columns** — the one cause that spans the whole postoperative course.
- The trap it sets: assuming a late obstruction must be a stricture. **Stones and lymphocele also present late.**

Diagnosis

- Suspect on **any deterioration of renal function, particularly with hydronephrosis of the transplant**.

- **Hydronephrosis may be less than expected**, because surrounding fibrosis stops the system expanding — hence **antegrade nephrostogram or a Whitaker test**.

Treatment

- The priority is **decompression, to let the allograft recover** — nephrostomy, nephroureteral stent, antegrade or retrograde stent.
- **Antegrade access is preferred**: retrograde insertion is hampered by neo-orifice position, stricture location and tortuosity, while antegrade nephrostomy gives a detailed nephrostogram and versatile instrument access. **Once a stent is in, retrograde exchange is straightforward**, so long-term nephrostomy is unnecessary.
- Endoscopic versus open depends on **performance status, comorbidity, and stricture length and location**.
- Endoscopic ranking: **antegrade double-J alone performs poorly; balloon dilation is the most studied and moderately durable; endoureterotomy does better** — holmium laser ahead of the Acucise cutting balloon, and both ahead of the resectoscope T-loop.
- **Open repair remains the most effective modality, with superior long-term results**. Options by size and site: allograft ureteric reimplantation, ureteroureterostomy to the ipsilateral native ureter, Boari flap, appendiceal or ileal interposition, pyelovesicostomy.
- **Robot-assisted translation of these repairs gives comparable short-term outcomes**; indocyanine green identifies the ureter and assesses viability.

KEY POINTS: URETERAL STRICTURE

- Strictures are the predominant cause of ureteral obstruction beyond 3 months after transplantation.
- The primary cause of stricture development is ischaemia.
- Stricture should be suspected with hydronephrosis and decreased graft function.
- Endourologic options exist, but open reconstructive surgery offers superior long-term results.

VESICoureTERAL REFLUX

- Several anastomotic modifications exist to replicate an antireflux mechanism, yet **reflux is demonstrable on voiding cystography in 50% to 86%, while clinically significant reflux occurs in under 1%** — and that figure is **independent of anastomotic technique**. The modifications therefore change the radiology, not the clinical problem.
- Sequelae are unclear: one study found **more hypertension at 5 years and a trend to urosepsis**.
- Causes: **technical error at ureteroneocystostomy, bladder outlet obstruction, or high-pressure storage** from detrusor overactivity or poor compliance.

Diagnosis

- **Recurrent UTI prompts voiding cystography**; once reflux is confirmed, **exclude bladder outlet obstruction and high-pressure storage**.

Treatment

- Therapy targets **the underlying cause, not the reflux**, in most cases; outlet obstruction and high-pressure storage are treated conventionally.
- Traditional surgery is **revision of the anastomosis or reimplantation into the ipsilateral native ureter**.

- **Endoscopic submucosal injection (Teflon or Macroplastique) succeeds in low-grade and fails in high-grade reflux**, so it is confined to clinically significant grade I to II disease.
- **High-grade reflux, or failed endoscopy with persistent symptoms, goes to reconstructive surgery.**

LYMPHOCELE

- A pseudocyst of lymph within a hard fibrous capsule around the graft.
- Radiographically evident in 0.6% to 33.9%; symptomatic in 0.03% to 26%; most develop within 6 months, **peaking at 6 weeks**.
- Mechanism: **extensive dissection of lymphatics around the recipient iliac vessels or the donor renal vessels**, at procurement or on the back table.
- Associations: obesity, recipient age, dialysis duration, warm ischaemia time, **prophylactic low-molecular-weight heparin**, delayed graft function, acute rejection, re-transplantation, **mammalian target of rapamycin (mTOR) inhibitors**.
- Most are silent. Large ones compress: **unilateral limb oedema, graft dysfunction, bladder compression, fever, deep vein thrombosis from external iliac vein compression**.

Diagnosis

- Ultrasound or CT shows the collection; **biochemical and microbiologic analysis of drain fluid or a guided aspirate separates lymphocele from urinoma, seroma and abscess**.

Treatment

- Most resolve spontaneously: **fewer than 15% need treatment**.
- Efficacy runs **aspiration alone worst, drain placement and sclerotherapy intermediate, laparoscopic decortication with peritoneal window best, open surgery close behind**.
- **Open drainage for wound complications, and for a small lymphocele adjacent to vital renal structures** — the two settings where laparoscopy is unsafe.

NEPHROLITHIASIS

- Incidence about 1%, mean onset 28 ± 22 months; **calcium-based commonest, then struvite, then uric acid**.
- Lithogenic factors: **secondary hyperparathyroidism, recurrent UTI, and metabolic abnormality** — hypercalciuria, hyperuricosuria, hypocitraturia, hyperoxaluria.
- Drug effects on the urinary environment: **ciclosporin causes chronic hyperuricaemia; calcineurin inhibitors cause hyperoxaluria and hypocitraturia**.

Treatment

- **Follows the native-kidney algorithm**. Asymptomatic non-obstructing stones **under 4 mm** may be observed with serial ultrasound and creatinine.
- Growing or symptomatic: **extracorporeal shock wave lithotripsy, retrograde intrarenal surgery, percutaneous nephrolithotomy, or combinations**.
- Afterwards: **watch for silent hydronephrosis following instrumentation** — denervation again removes the warning — and **screen metabolically**.

URINARY TRACT INFECTIONS

- **The commonest infectious complication**, causing 30% of sepsis admissions; prevalence 7% to 80%, highest in the first 6 months.
- **No sex difference immediately after transplantation**, but a significantly higher rate in females by 3 years.
- Transplant-specific risks: **prolonged stent, reflux, rejection, poor graft function, high prednisolone dose**.
- Definitions are unchanged from non-transplant practice: asymptomatic bacteriuria; simple cystitis; pyelonephritis or complicated UTI (fever, chills, haemodynamic change, allograft pain, or bacteraemia with the urinary pathogen); recurrent UTI as **three or more in 12 months**.
- **Screening has not been shown to reduce complications beyond 2 months post-transplant**, and **treating asymptomatic bacteriuria is poorly supported** — except where creatinine has risen unexplained.
- Acute cystitis: empirical **fluoroquinolone, amoxicillin-clavulanate, third-generation cephalosporin or nitrofurantoin (creatinine clearance above 30 mL/min)**, then tailored. 5 to 10 days; avoid shorter courses.
- Complicated UTI or pyelonephritis: **14 to 21 days**, allowing for **greater drug resistance in recipients**, and **confirm the kidney is draining**.
- Recurrent UTI demands **structural evaluation** — **anatomy, reflux, bladder dysfunction, outflow obstruction**.
- Non-antibiotic measures are poorly studied here; **vaginal oestrogen is recommended for postmenopausal recipients by extrapolation**.

LOWER URINARY TRACT COMPLICATIONS

Voiding Dysfunction

- Common after transplantation and **concealed before it**: oliguria masks outlet obstruction and incontinence, so the problem only declares itself once urine output returns.
- Three interacting contributors: **small bladder capacity (under 300 mL), pre-existing anatomic abnormality, and a direct effect of immunosuppression reducing bladder mass**.
- Evaluation: **urologic and neurologic history, previous urologic surgery, full urologic and gynaecologic examination**, with **International Continence Society questionnaires, the International Prostate Symptom Score, and a voiding diary**. Uroflowmetry, urodynamics and cystoscopy are used selectively by risk and symptoms.

Small Bladder After Transplant

- **Capacity in anuric recipients grows back to normal after transplantation**, and the consequence is explicit: **pretransplant bladder rehabilitation or surgical augmentation is unnecessary**.
- Where symptoms affect quality of life: **anticholinergics, then botulinum toxin** for insufficient capacity or high storage pressure; **augmentation or ileal conduit for refractory cases**.

Benign Prostatic Hyperplasia

- An age-related disease, therefore **commoner in recipients**, and **hard to detect in anuric patients**. **Three-year incidence 9.7%**, associated with **increased UTI and graft loss**.

- Where significant obstruction is suspected: **α -blocker, aggressive treatment of constipation before catheter removal**, and in selected cases **a suprapubic tube at transplantation**.
- **Resecting the prostate before transplantation, while anuric or oliguric, risks bladder neck contracture and urethral stricture** — healing without urine flow is the mechanism.
- **Transurethral resection or holmium enucleation is safe and effective at 3 weeks or more after transplantation.**
- **Avoid transurethral resection in the first 2 weeks and with a ureteral stent in place** — sepsis and death have followed.

Urinary Incontinence

- Prevalence among female recipients **28%**.
- **Stress incontinence is diagnosed and treated as in non-transplant patients:** tension-free vaginal tape, transobturator tape, sacral neuromodulation and laparoscopic sacrocolpopexy are all reported as feasible and safe.
- **An artificial urinary sphincter after radical prostatectomy is feasible** in recipients, on limited evidence.

ERECTILE DYSFUNCTION AND HYPOGONADISM

- **Erectile dysfunction reaches 87.5% of men with ESRD. Transplantation improves it, yet 20% to 50% remain affected**, with only a modest gain in International Index of Erectile Function score and **no significant improvement where pretransplant dysfunction was severe.**
- The mechanism is unclear; **earlier transplantation does better, probably by pre-empting penile vasculopathy.**
- **Treatment is identical to non-transplant practice. Hypogonadism occurs in ESRD and testosterone improves after transplantation.**

GENITOURINARY MALIGNANCIES

Renal Cell Carcinoma

- **Incidence exceeds the general population** — 4.6% of post-transplant cancers against 3% of tumours generally.
- **Only 10% arise in the graft; the rest are in the native kidneys.** Mean time to a de novo graft tumour is 131 months.
- Graft tumours are favourable: **papillary histology predominates, and most non-papillary graft tumours are low grade (Fuhrman 1 and 2).**
- **Treatment mirrors the general population. No recommendation exists on modifying immunosuppression after a cancer diagnosis**, though for metastatic graft-origin disease some propose **stopping immunosuppression after allograft nephrectomy**, so the recipient rejects the donor-derived tumour cells.
- **Kidney Disease: Improving Global Outcomes (KDIGO) does not endorse screening the native kidneys**, benefit being unproven against harm. **Consider screening high-risk recipients only:** previous renal cell carcinoma, analgesic nephropathy, tuberous sclerosis, acquired cystic disease.

Prostate Cancer

- **Incidence matches the general population**, though prostate-specific antigen (PSA) testing does not: **PSA is generally part of the older male recipient's pre-transplant workup**, yet **no national recommendation exists for or against screening recipients** beyond general-population guidance.
- Management is constrained by **the graft's anatomic proximity to the surgical or radiation field**, and by immunosuppression raising **infection, lymphocele, impaired healing and wound complications**.
- Most treated recipients have **organ-confined disease, Gleason score 6 or less. Surgical extirpation is commonest, then radiotherapy, then active surveillance — which is underused.**
- **Oncologic outcomes across modalities match non-transplant patients.**

Bladder Cancer

- **Risk is roughly threefold** (standardised incidence ratio 3.18; $P = .008$), and the disease is **more aggressive — higher recurrence, progression and metastasis**, so **aggressive surveillance is necessary**. Mean interval from transplantation to diagnosis **2.8 to 4 years**.
- **Intravesical bacillus Calmette-Guérin remains usable in selected high-risk non-muscle-invasive disease** despite immunosuppression.
- **Muscle-invasive disease: chemoradiotherapy, or radical cystoprostatectomy with urinary diversion.** At cystectomy and node dissection, **protect the graft vessels, the ureteric blood supply and the ureter.**

KEY POINTS: GENITOURINARY MALIGNANCIES

- Most kidney tumours develop in the native kidneys and are low grade.
- Active surveillance, radiation and surgery are all reasonable for prostate cancer in recipients.
- Recipients carry a threefold risk of urothelial carcinoma, and it is biologically more aggressive.

CONCLUSION

- Recipients face multiple urologic complications, but **management largely mirrors the general population, with defined special considerations.**
- **Diligence in recognition and treatment is what secures survival of graft and host.**

FIGURE AND TABLE LEGENDS

- **Box 92.1 Additional Causes of Urinary Leak** — reproduced in full above.
- **Table 92.1 Common Causes of Ureteral Obstruction** — reproduced in full above.
- **Fig. 92.2 Proximal ureteral stricture (arrow).** Antegrade nephrostogram reveals hydronephrosis in the allograft without visualisation of the ureter.
- **Fig. 92.3 Renal mass in a transplanted kidney.** CT reveals an exophytic renal mass (arrow) in a non-functioning allograft in an 80-year-old patient. Biopsy showed low-grade papillary renal cell carcinoma type 1; cryotherapy was used to ablate the mass.

Urologic Considerations in Paediatric Renal Transplantation

- The defining contrast with adults: **children needing kidney replacement therapy (KRT) usually have a congenital anomaly of the kidney and urinary tract (CAKUT), including obstructive uropathy** — not hypertension and diabetes.
- The paediatric urologist's task is to **ensure the condition that damaged the native kidneys does not go on to damage the transplant**.
- Children and adolescents need a **multidisciplinary team, ideally with urological involvement before the onset of end-stage kidney disease (ESKD)**.
- Five goals: unimpeded drainage from kidney to lower tract; a **capacious, low-pressure reservoir holding enough for socially acceptable continence**; volitional emptying; absence of infection; the fewest surgical procedures.
- **A reactive approach — referral only after an allograft is in and already in jeopardy — is suboptimal and should be avoided.**
- Early lower tract optimisation makes transplantation smooth; **pre- and postsurgical assessment and monitoring drive early detection of complications and good long-term outcomes.**
- Loss of function from chronic rejection is unavoidable, but **every additional insult accelerates it, and diligent post-transplant care limits that.**

PRETRANSPLANT ASSESSMENT

- An important proportion of children needing KRT have **uropathy: congenital obstruction, vesicoureteral reflux (VUR), or renal dysplasia, hypoplasia or aplasia.**
- **Commonest age at paediatric transplantation is 13 to 17 years.** Commonest congenital conditions are **hypoplasia, aplasia and dysplasia, and obstructive uropathy**; the commonest acquired cause is **focal segmental glomerulosclerosis**. Adult series are led instead by hypertension and diabetes, which rarely cause ESKD in the young.
- The age split: **younger children have dysplasia or obstructive uropathy — chiefly posterior urethral valves (PUVs) in boys — while nephrologic causes and neuropathic bladder dysfunction present in older children, adolescents and young adults.**
- **Screen all children before transplantation**, accepting a low yield where the cause is clearly nephrologic. **History, renal ultrasound, and uroflow with postvoid residual (if toilet trained) exclude most abnormalities.**
- **Routine voiding cystourethrography (VCUG) is not necessary.** Consider it for neuropathic bladder, obstructive uropathy, VUR, prior lower tract surgery, recurrent febrile urinary tract infections (UTIs), unexplained worsening hydroureteronephrosis, or difficult-to-treat voiding abnormalities.

Focused Assessment

- Assessment is **directed by the underlying condition and the status of bladder and kidneys.**
- **VCUG shows anatomic change with filling and emptying, the urethra, and the presence and grade of reflux.** Abnormal dynamics on any test justify urodynamics.
- **Indications for urodynamics:** neuropathic bladder; outlet obstruction from PUVs or a large ureterocele; and selected cases with recalcitrant voiding dysfunction, high-grade hydronephrosis, or recurrent UTI.

- Urodynamics assess **capacity, compliance, detrusor activity, sensation, emptying and sphincter function** — to **characterise dynamics that place the allograft at risk** and define the therapy needed.
- Therapy that follows: **antimuscarinics, α -blockers or β 3-adrenoceptor agonists; clean intermittent catheterisation (CIC); intradetrusor botulinum toxin; lower tract reconstruction including augmentation.**
- **Emptying matters as much as storage, because it governs infection risk** — and **treating poor compliance medically can itself impair emptying and force CIC.**
- Where the urethra is **sensate, or difficult to access or negotiate, a continent catheterisable stoma is the best route to effective CIC.**
- **Urodynamics are never a one-off:** repeat testing tracks response and change over time.

KEY POINTS: UROLOGIC EVALUATION AND MANAGEMENT

- Goals: low-pressure, unimpeded drainage into the bladder or reservoir; low-pressure storage for a socially acceptable time; volitional emptying without incontinence; absence of infection; fewest procedures and least trauma.
- A large fraction of children needing KRT have uropathy — congenital obstruction, VUR, or renal dysplasia.
- Pretransplant assessment is directed by the underlying condition and the status of bladder and kidneys.
- Urodynamics assess capacity, compliance, emptying and sphincter function.
- Indications include neuropathic bladder; a history of PUVs; and recalcitrant voiding dysfunction, high-grade hydronephrosis or recurrent UTI.

PRETRANSPLANT PREPARATION

- **The commonest early bladder abnormalities in ESKD are low capacity, detrusor overactivity, and poor compliance.**
- This is the typical initial picture in PUVs, but **bladder dynamics evolve:** from increased contractility, through amelioration with better capacity, to **insufficient contractility and inefficient emptying — myogenic failure.**
- **Overactivity and poor compliance are the dangerous patterns, because high pressure reaches the kidneys even without reflux.** Failure to address them puts the allograft at risk.

Detrusor Overactivity, Low Capacity and Poor Compliance

- The goal is to **store up to estimated capacity for age at detrusor pressures below 40 cm H₂O during filling or at leakage, with social continence.**
- **Initial management is medical — antimuscarinics. Surgery, augmentation included, is second line.**
- Medical management often needs **CIC**; children may try to avoid it by double voiding or Valsalva emptying, and **α -blockers may aid emptying.**
- **Never assume volitional emptying is effective** — demonstrate it with **repeated low postvoid residuals and uroflowmetry.**
- **Oxybutynin is titrated upward against side effects,** watching catheterised volumes, incontinence and hydronephrosis. Tolerability improves with **selective antimuscarinics (solifenacin, darifenacin), β 3 agonists (mirabegron), or intradetrusor botulinum toxin.**

- **Compliance with pretransplant CIC is an excellent test of adherence** to the demands of transplantation; a family that cannot manage CIC may not manage the rest of post-transplant care.

Capacity

- There has been a **trend away from augmentation toward less invasive medical management** — perhaps to the extreme. Augmentation remains valuable in selected failures of medical management.
- **Current indications:** a small, non-compliant, high-pressure bladder, and failure of medical therapy plus CIC to achieve low-pressure storage for up to 3 to 4 hours. Where pressures are safe only briefly, or regularly exceed 40 cm H₂O despite maximal therapy and catheterisation, augment.
- **No compelling evidence that augmentation increases complications during or after transplantation.**
- Segment choice depends on **anatomy, prior surgery, and whether a concurrent continent catheterisable stoma is planned.**
- **Gastric augmentation** was popular in the late 1980s and early 1990s on the theoretical grounds of **acid secretion in an acidotic patient, less infection, fewer bladder stones**, but is **now rarely offered**: it causes severe complications, **especially in the pretransplant anuric group**, through gastric secretion injuring the native bladder and producing the **haematuria-dysuria syndrome**.
- **Ileum and sigmoid are the commonest segments.** Against them: **reports of malignancy after augmentation, particularly in the transplant group** — **long-term surveillance is essential in all cases.**
- The augment patch rarely obstructs the transplant, but the pedicle's presence, location and orientation must be recognised.
- **Continent stomas must be sited clear of the anticipated transplant incision, and nearly always lie medial to it.**
- **Most allografts go retroperitoneally through a Gibson-type incision**, so intraperitoneal dissection is rarely needed and stomal position rarely matters. **In infants and small children under 15 kg the vascular anastomosis may have to be fashioned at the aorta and vena cava, requiring a transperitoneal approach** — though such children rarely have an augment or channel in place.

Infections

- Many of these children have recurrent UTIs. **Graft pyelonephritis causes major morbidity in an immunocompromised host and can end in graft loss.**
- The usual cause is **urinary stasis** — inadequate emptying by voiding or catheterisation, persistent obstruction, or VUR.
- **UTIs do not improve after transplantation, so recurrent infection must be resolved in advance: double and triple voiding, or CIC, for timely complete emptying.**
- **Differentiate true UTI from colonisation, which is exceedingly common on CIC.**
- **Persistent UTI despite adequate emptying and good CIC technique means an anatomic abnormality or chronic colonisation.** Colonisation may respond to **long-term antibiotics, intravesical antimicrobials, or regular bladder irrigation**; the anatomic culprits — **diverticula and ureteral stumps** — need targeted imaging and surgery.

The Defunctionalised Bladder

- A bladder that **lacks cycling because of anuria or supravescical diversion.** The difficulty is that its **response to physiologic urine volumes cannot be predicted**, and this is compounded by any associated lower tract dysfunction.

Neuropathic bladder.

- **Becoming rare with better management**, but still seen where urologic care has been suboptimal. Potential function is hard to predict, and a **defunctionalised bladder takes time to reach its maximal potential**.
- Rehabilitation is by **cycling**, as a sequence: catheterise → instil progressively increasing volumes of saline → hold for a set dwell time → drain → increase volumes at regular intervals, **empirically, from the initially tolerated volume**.
- The response indicates **the safety of the native bladder as a reservoir and any ability to empty spontaneously; adjunctive medication is often needed** to increase capacity and compliance.
- **The target is expected capacity for age** by any standard formula. Goal volumes may not come immediately, but **steady increases without discomfort or significant leakage predict further expansion**.
- **Safe storage pressures, capacity and compliance must be confirmed formally by urodynamics, and only then is the child cleared for transplant**.
- **A renal allograft must never be used to test whether a bladder works**. If adequate storage and compliance cannot be reached, augmentation must be considered.

Non-neuropathic bladder.

- Seen with **PUVs or VUR**, and in **anuric children without prior urologic problems** — including those after bilateral nephrectomy.
- **These bladders can often be salvaged for storage and spontaneous voiding, but are approached exactly as the neuropathic one**, assuming CIC may be needed long term.
- **Spontaneous emptying determines whether a continent stoma is needed** — more pressing in children with normal sensation, who resist urethral catheterisation.

Decision to augment.

- **Early proactive bladder management reduces or delays the need to augment**. The strategy that identifies real candidates is **regular CIC, overnight bladder drainage, and maximised medical therapy**.
- **Enterocystoplasty remains one of the most effective means of providing adequate storage in advanced lower tract dysfunction**.
- **Broad indications before transplantation**: inability to protect the upper tracts and reach **functional capacity above 75% of expected for age**, at **pressures below 40 cm H₂O**, while **catheterising no more often than every 3 to 4 hours**. The authors concede these **thresholds are somewhat arbitrary and lack strong data**.
- **Timing is condition-specific**: in PUVs, progression to myogenic failure makes it prudent to postpone augmentation as long as possible; a neuropathic bladder with borderline urodynamics despite maximal therapy may deteriorate rapidly under a robust post-transplant urine output, arguing for earlier reconstruction.
- **Complications of enterocystoplasty are time-dependent and accumulate across a transplanted child's lifespan**, compounded by an immunocompromised state that impairs infection control and healing and raises malignancy risk. **Recommending it is a major lifelong commitment and an individualised decision**.
- **Always explore a dilated ureter for ureterocystoplasty**. Outcomes are mixed and depend on ureteral diameter, bladder compliance and VUR, but it is **preferable to enterocystoplasty on mucus, stones, infection and metabolic grounds**.

- Hence **preserve the ureter** — for augmentation, for a continent catheterisable channel, or as the target for allograft ureteral anastomosis.

Cutaneous Stomas

- The defunctionalised bladder may coexist with **ileal and colon conduits, vesicostomies, pyelostomies or ureterostomies**, and the donor ureter can be drained into these diversions.
- **Contemporary management centres on definitive bladder optimisation before transplantation.**
- **Long-term stoma or transplant ureterostomy is reserved for severe neurodevelopmental compromise, or inability to tolerate CIC.**
- **A functioning ileal loop can serve as the enterocystoplasty patch during refunctionalisation, even in an anuric patient.** If augmentation proves unnecessary and the child still makes urine, **implant the native kidney(s) to refunctionalise the bladder and discard the loop.**

KEY POINTS: PRETRANSPLANT PREPARATION

- The commonest bladder abnormality with uropathy and ESKD is a low-capacity, high-pressure bladder with poor compliance.
- No clear evidence that bladder augmentation increases the risk of transplant.
- Recurrent pyelonephritis is a hazard for the transplant and is associated with graft loss.
- Initiating CIC before transplantation serves assessment as well as emptying.
- Refunctionalisation is best achieved by cycling, to increase capacity, determine compliance, and assess the family's ability to perform CIC.

RECONSTRUCTION STRATEGIES

- Factors to weigh before reconstructing: **dialysis modality, likely graft location, timing relative to KRT, need for bowel surgery, status of the native kidneys.**

Dialysis Issues

- In a child on peritoneal dialysis (PD), intraperitoneal surgery will usually force temporary or permanent transition to haemodialysis.
- An intraperitoneal procedure can sometimes be done at **PD catheter placement**, but **adding bowel surgery raises the risk of catheter infection and removal.** Preserving peritoneal integrity is always **desirable**, PD being the best temporary KRT in children.
- Where KRT is approaching, **starting dialysis before reconstruction improves metabolic status** — promoting healing, reducing complications, and helping the child tolerate the metabolic load of urine reabsorption where bowel is used.
- Against that: **extensive intra-abdominal surgery may preclude peritoneal use**, and on haemodialysis there is risk of **catheter infection with postoperative sepsis, and arteriovenous fistula injury from low flow or direct pressure.**

Graft Placement

- **Anticipate the future allograft site.** In the very small child, where the graft may lie intraperitoneally, **move mesenteric pedicles away from the midline.**
- Very small children rarely have had prior reconstruction, and **most allografts are still accommodated extraperitoneally.**

- **A prior psoas hitch, or extensive dissection for native ureteral reimplantation, can make ipsilateral iliac graft placement difficult.**
- Where alternatives are few, **documentation and collaboration with the transplant team are essential:** a detailed operative report and a diagram are of great value to future surgeons.

Timing

- **Perform major urologic reconstruction before transplantation.**
- **Allow 6 to 12 weeks for healing after enterocystoplasty or continent diversion.**
- **Unilateral native nephrectomy at transplantation is reasonable, particularly ipsilateral to the graft site.**
- **Avoid concurrent procedures in infants and small children** — worse outcomes with prolonged anaesthesia and haemodynamic instability, and **prophylactic anticoagulation against graft thrombosis adds bleeding risk.**

Enterocystoplasty

- **Feasible in a dialysis patient or a child approaching ESKD, and done as in any other child.**
- Segment and strategy depend on **need for a continent stoma, availability of native bladder tissue, the bladder neck and continence mechanism, and bowel function.**
- Where continent diversion is needed for an unsalvageable bladder, **there is no reason to defer it until after transplant** — **except in the child transplanted in the first 2 years of life, before continence is expected or beneficial.**
- **The often-quoted metabolic problems of continent diversion in renal impairment are not an absolute contraindication,** though careful medical management is needed afterwards.
- Two directions to hold together: **in advanced chronic kidney disease considered for pre-emptive transplantation, urine reabsorption through bowel may accelerate the need for dialysis; in less marginal cases, better bladder dynamics and less infection may delay progression to ESKD.**

Native Nephrectomy

- **The decision is multidisciplinary and controversial; the default is to leave the native kidneys.**
- **Indications for removal before transplantation:** difficult-to-control hypertension; profound nephrotic syndrome with malnutrition or hypercoagulability from protein loss; massive polyuria; recurrent upper tract infection; large stone burden; high malignancy risk; very large kidneys (for allograft space or to allow PD); massive VUR.
- **Massive VUR is relative** — but with stasis and infection in an immunosuppressed child, **removal may be safer than reimplantation or conservative management.**
- **The approach is decided by whether the child is on PD.**
 - On PD, **entering the peritoneum, open or laparoscopically, requires temporary haemodialysis, and open surgery may scar the peritoneal membrane and make later PD ineffective.**
 - **Retroperitoneal nephrectomy allows immediate PD,** though leaks occur if the peritoneum is breached.
 - For **bilateral nephrectomy, the prone approach avoids repositioning and is optimal with relatively small kidneys.**
 - On PD with large kidneys: **posterior open nephrectomy, or laparoscopic nephrectomy with morcellation.**

- Not on PD: **any approach is acceptable**; laparoscopic adhesions are limited and unlikely to compromise future PD, and **the catheter can be placed at the same procedure if PD is imminent**.
- In selected high-risk cases, **renal embolisation is a reasonable alternative**.
- The case for leaving them: **water excretion while on dialysis or if the graft fails, reducing fluid restriction; erythropoietin production and vitamin D metabolism; and avoidance of nephrectomy complications**. Most allografts are transplanted without removal, particularly if the kidneys are small or atrophic.

Limiting the Risk of Infection

- **Prevention must be proactive, not a response to systemic compromise** — additive damage to the allograft determines long-term outcome.
- Situations warranting active prevention: **high-grade reflux; persisting hydronephrosis with or without reflux; dependence on intermittent catheterisation**, that group being frequently colonised and more susceptible.
- **The non-dilated, non-refluxing native kidney is unlikely to be the source of infection**.
- Risk is multifactorial: **discuss the protective value of circumcision with families of boys at risk of ESKD, particularly those with valves**, considering it early to delay progression to chronic kidney disease and reduce post-transplant infection.

Preserving the Ureter(s)

- At nephrectomy, **transect the ureter as high as possible** to minimise adhesions and fibrosis around the iliac vessels.
- **A normal ureter should always be left** — it limits dissection near the iliac vessels and preserves **primary or salvage ureteroureterostomy for distal transplant ureteral stenosis**.
- **A dilated ureter with high-grade reflux or obstruction should be removed entirely, to limit infection risk**.
- If bladder function is abnormal and catheterisation may be needed, **preserve the ureter as a possible continent stoma** — best done pretransplant, though creation at transplantation remains an option.

Combining Nephrectomy and Transplantation

- **Generally avoided**, to limit surgical morbidity and anaesthetic time.
- Done mainly **to create space** — **small children receiving an adult allograft, or very large native kidneys** — and is **most appropriate as a single ipsilateral nephrectomy through the transplant incision**.

KEY POINTS: PRETRANSPLANT RECONSTRUCTION

- The ability to empty the bladder spontaneously determines the need for a continent catheterisable stoma.
- Indications for augmentation before transplantation: capacity less than 75% of expected for age; detrusor pressures above 40 cm H₂O; catheterisation every 3 hours or less; maximal medical therapy.
- For patients on PD, intraperitoneal surgery will likely require temporary transition to haemodialysis.
- Any major urologic reconstruction should be undertaken well before anticipated transplantation.
- Native nephrectomy is indicated for: difficult-to-control hypertension despite multiple medications; profound nephrotic syndrome with malnutrition from protein loss; high risk for malignancy;

recurrent upper tract infections; large kidneys, to make space for the transplant; large stone burden; massive VUR.

TRANSPLANTATION

- The vascular component follows adult principles unchanged.
- **For the paediatric urologist not performing the vascular anastomosis, the lower urinary tract reconstruction is the focus of intraoperative attention and of urologic complications.**

Ureteral Anastomosis

Surgical Techniques and Options

- As with reimplantation for VUR, **intravesical and extravesical techniques both exist.**
- The authors **favour a non-refluxing anastomosis in most cases**, acknowledging divergent views and that an antireflux mechanism is not always essential. The argument: **screening identifies bladder dysfunction and therefore graft-reflux risk, and a non-refluxing reimplantation is simple, effective and of minimal morbidity.**
- **Extravesical ureteroneocystostomy is preferred**, as a sequence: after reperfusion and haemostasis, **partially fill the bladder with saline or dilute antibiotic solution** → clear the anterolateral aspect and place **traction sutures** to mobilise it upward and tension the wall → **size the ureter to ensure it reaches** → **incise detrusor to mucosa over 3 to 4 cm in an anatomic direction, avoiding kinks** → **elevate detrusor flaps off the mucosa and excise a small disc of mucosa at the distal trough** → drain the bladder, traction sutures holding the trough → **trim and spatulate the ureter for 4 to 5 mm** → **interrupted mucosa-to-mucosa anastomosis with fine absorbable sutures** → **stitches through detrusor and terminal ureteral adventitia to prevent eversion** → **close the detrusor flaps over the ureter, not tightening the entry point.**
- **Alternative — the Barry technique:** a 4 cm tunnel between parallel incisions through which the ureter is passed. **A shorter tunnel has been used in some paediatric centres with acceptable outcomes.**
- **Ureteroureterostomy to the native ureter is routine in some centres with good results**, done **side-to-side, end-to-side or end-to-end**. With the native kidney in place, some **ligate the proximal ureteral segment**, without major complications and with few later nephrectomies.
- **Superiority over ureteroneocystostomy is unproven**, but it appeals with **a normal non-refluxing native ureter, a small bladder, a short donor ureter, or difficult pelvic dissection after previous surgery**, and should always be considered as salvage for post-transplant ureteral stenosis.

Ureteral Stenting

- **Routine stenting in paediatric transplantation is highly debated, with no conclusive data for routine use.**
- **Stent for a difficult reimplant — particularly into an abnormal bladder — or a traumatised or devascularised graft ureter.**
- **Adult series show stenting reduces ureteral complications from 5.8% to 1.9%, at a higher risk of infection — the trade-off in one line.**
- **If used: a short double-J stent, removed at 4 to 6 weeks;** some advocate **early removal at 5 to 7 days**, securing the stent to the tip of the urethral catheter or using an externalised stent or splint, which **may reduce stent-related complications and infection.**
- **Removing an internal stent needs cystoscopy**, and can be coordinated with removal of tunnelled intravenous access or a PD catheter.

KEY POINTS: TRANSPLANT URETERONEOCYSTOSTOMY

- Extravesical non-refluxing ureteroneocystostomy is the preferred ureteral anastomosis.
- Transplant to native ureteroureterostomy is an effective option.
- There are no data to support routine ureteral stenting at transplantation.
- Early stent removal is not associated with increased complications and may reduce infection.

COMPLICATIONS

- **The incidence of complications is low and rarely costs the allograft.**
- **Thrombosis is the paediatric-specific worry — one of the commonest causes of early graft loss**, which is why some centres anticoagulate routinely early, **accepting more bleeding during reconstruction and demanding meticulous haemostasis.**
- Comparison of recipients with and without prior obstructive uropathy or reflux shows **an increased risk of all urologic complications where uropathy pre-exists.**
- **PUVs have been linked to increased graft dysfunction in some series but not others**; obstructive uropathy carries a higher complication risk in many. The practical conclusion: **special vigilance and a lower threshold for intervention.**

Urine Leaks

- **Rare — under 5% — and identified early**, with wound drainage, increased perirenal drain output, or a perinephric collection on ultrasound.
- **Fluid creatinine distinguishes urine from lymph or peritoneal fluid**, relevant after a transperitoneal approach or inadvertent peritoneal opening.
- **Assessment as a sequence**: check all urinary drainage tubes, **especially the bladder catheter** → if in place, **irrigate and confirm patency**; if removed, **replace it** → **transplant ultrasound for hydronephrosis**; if present, **suspect distal ureteral obstruction and consider percutaneous nephrostomy** → **document stent position** → **localise the leak with MAG3 (if graft function allows) or cystogram**, the latter showing a bladder leak at the anastomosis, or a pyelocalyceal leak where a stent is in place and contrast refluxes.
- **Indications for intervention are clinical: a limited leak may be observed.**
- **In young children with very high urine output, the leak may be driven by poor drainage through a small bladder catheter**; upsizing can resolve it. Hence the preventive rule: **the catheter placed at transplantation should be the largest the child's urethra accepts.**
- **Persistent worsening or a large leak despite adequate bladder drainage: percutaneous renal drainage and/or exploration.**
- Exploration aims to **identify the cause, confirm the site, and repair**:
 - **Bladder-based leak** — revise the anastomosis, close the defect, drain adequately.
 - **Distal ureteral necrosis** — replace the ureter. Short segment: bladder mobilisation and reimplantation. Long segment: native ureter, ipsilateral or contralateral. No native ureter: psoas hitch or bladder flap.
- **These are the same strategies as for ureteral strictures; careful stenting and drainage are essential.**

Infection

- **Early infections relate to the indwelling catheter, the stent, and instrumentation.**

- **Late urinary infection is a relatively frequent delayed complication reflecting bladder function and the underlying urologic cause of renal failure in an immunocompromised host.**
- **High-grade or worsening hydronephrosis may herald pyelonephritis and worsening function, and needs prompt evaluation.**
- **Surveillance: routine assessment of bladder emptying, detection of hydronephrosis, selective VCUG.**
- **On CIC with no correctable cause: prophylactic antibiotics and bladder irrigation may work. Proactive management of bladder dysfunction, identified pretransplant, is essential.**
- **Graft VUR with documented recurrent febrile infection warrants surgical correction; reflux without infection and with normal bladder function may be observed.**

Reflux

- **VUR into a transplant is entirely distinct from reflux into a native unit, for three reasons: it is a reimplanted ureter; pyelonephritis threatens a transplanted kidney more; and the patient is immunosuppressed.**
- **Routine post-transplant evaluation is controversial but performed in some centres for risk stratification. Febrile UTI is a clear indication for cystogram.**
- **With optimised bladder dynamics and recurrent febrile UTIs, transplant reflux justifies open or robotic reimplantation. Uncorrected, allograft pyelonephritis recurs and graft loss risk rises.**
- **Identify simultaneous bladder dysfunction, though it may not be correctable. Asymptomatic reflux — no infection, no bladder dysfunction — should be observed.**
- **Surgical management of transplant reflux is never routine.**
- **Endoscopic therapy has a limited role: resolution rates are modest to poor, and post-injection obstruction is commoner than in other indications, with significant morbidity. It suits selected low-grade cases; recurrent febrile UTI justifies definitive intervention.**
- **Data on intravesical versus extravesical reimplantation are few; the authors prefer intravesical, often with extravesical mobilisation. Transtrigonal if the contralateral native ureter can be avoided; otherwise advancement or Politano-Leadbetter. Stenting is advisable.**
- **Where the transplant ureter was joined to the ipsilateral native ureter, options are similar but response to injection and ease of dissection may be better, the patient's own ureterovesical junction being preserved.**

Hydronephrosis and Obstruction

- **Hydronephrosis is frequent and indicates ureteral obstruction in as many as 8% of transplants.**
- **The transplant kidney is particularly sensitive to obstruction, and the degree of impairment does not always correlate with the grade of hydronephrosis. High-grade or worsening dilatation prompts further evaluation.**
- **Obstruction is often heralded by rising creatinine and cystatin C.**
- **More than half of obstructions occur within the first 100 days — hence long-term monitoring with regular ultrasound.**
- **With normal prior bladder function, this pattern means ureteral obstruction until proven otherwise. Where bladder dysfunction is known or suspected, both are evaluated and managed together — obstruction is commoner with bladder abnormalities, particularly PUVs.**
- **An indwelling catheter is a reasonable first step: improvement in dilatation and function points to the bladder.**

- **Obstruction and rejection may be intermingled** when creatinine rises with hydronephrosis:
 - Mild hydronephrosis with other signs of rejection → biopsy first.
 - No suggestion of rejection and dilatation unimproved by bladder drainage → ureteral stent or nephrostomy, with or without biopsy.
 - Mild-to-moderate hydronephrosis, rising creatinine, rejection on biopsy, graft not failing rapidly → treat the rejection medically, reserving stenting for non-improvement.
 - Failed retrograde stenting → percutaneous drainage, then antegrade study to localise and antegrade stent in the same setting.
- Isolated VUR is unlikely to cause progressive hydronephrosis and worsening function.
- Diagnostic studies for obstruction are not completely reliable; keep a low threshold for stenting to test the effect on function. Rarely, an acutely failing transplant improves on decompression even without rejection.
- The cause is usually distal — stenosis at the reimplantation site — but may be anywhere along the ureter and often includes ischaemia toward the distal donor ureter.
- Neither implant method nor stent use has been conclusively linked to obstruction risk; bladder abnormality, particularly from PUVs, has.
- Extrinsic causes: lymphocele, or mass effect from post-transplant lymphoproliferative disease (PTLD).
- Definitive therapy follows the cause: focal narrowing on retrograde imaging → balloon dilation with stenting for 4 to 6 weeks; long-term stenting is used in adult series but of uncertain value in children; open definitive repair should not be excessively delayed, given the risk to the graft.
- Reconstruction uses native ureter or bladder flaps; avoid non-urothelial segments such as appendix where possible. Pyeloureteral anastomosis is an option if the native ureter is in place and healthy. All principles of reoperative reconstruction — well-vascularised functional tissue — apply.

Bladder Dysfunction

- Bladder dysfunction increases infection risk and impairs physiologic drainage, adversely affecting graft function.
- Distinguishing it from ureteral obstruction needs sequential diagnostic drainage: if a catheter passes easily, continuous drainage for 1 to 2 weeks with recheck of creatinine identifies bladder dysfunction when creatinine falls and hydronephrosis improves; if not, combine stent and bladder drainage and recheck.
- Treatment is behavioural and medical, plus instituting or intensifying CIC. Augmentation is rarely needed and only after maximal medical management. A continent stoma helps where urethral catheterisation is difficult.

Stones

- Uncommon — quoted up to 5%, more likely under 1% — yet dangerous for long-term graft survival. Across recipients of all ages the 3-year risk is about 2%, higher with a preoperative stone history.
- With appropriate management, stones do not increase graft loss, but are associated with more UTIs.
- Reported high incidences of bladder stones largely reflected retained suture material.
- Some stones are present in the allograft at retrieval; bench flexible ureteroscopy can remove small stones during cold preparation. Donor-gifted lithiasis is rarely obstructive or infective in adult data and can be safely observed; paediatric data are lacking.
- A new asymptomatic stone on surveillance prompts close monitoring and a search for a predisposing factor.

- **Intervene for stones over 1 cm, growing stones, or stones causing graft dysfunction from obstruction or infection.**
- **All conventional stone treatments are appropriate**, modulated by risk to the graft. **Choose the modality most likely to succeed at a single intervention, even if more complex** — weighing the **anatomic position of the transplant, stone size and location, and ease of retrograde ureteral access**, and individualising on presentation, imaging and renal function.

KEY POINTS: COMPLICATIONS

- Common complications of paediatric renal transplantation: ureteral stenosis (6%), urinary leak (3%), stones (2%), clinically significant reflux (2%).
- Urine leaks are identified early, with increasing fluid from the wound drains.
- Urinary infection is a long-term and often delayed complication largely reflecting bladder function.
- VUR into the transplant is entirely distinct from routine reflux, as the risk to renal function from acute pyelonephritis is greater.
- With a rising creatinine and hydronephrosis, obstruction and rejection may both be present.
- Ureteral obstruction is usually distal, with stenosis close to the reimplantation site.
- Bladder dysfunction may produce infection, but may also create an obstructive process impairing graft function.

SUMMARY

- Management of the paediatric ESKD patient with an underlying urologic condition requires **a thorough understanding of the patterns of bladder dysfunction and clear strategies for evaluation and treatment before transplant.**
- **Anticipating the needs and constraints of the transplant procedure, and being involved in all cases, gives a smoother procedure and continuity of care.**
- **Post-transplant monitoring must catch pathology before irreversible graft damage** — a high index of suspicion, and a clear sense of who is at high risk.
- **Multidisciplinary collaboration between paediatric nephrology, urology and the transplant surgical team is critical to maximise patient and graft survival.**

FIGURE LEGENDS

- **Fig. 61.20 Detection of leak at the site of ureteroureterostomy during cystogram**, with retrograde flow of contrast around a stent (arrow). Note reflux into the native ureter (arrowhead) up to the renal pelvis, and a drain (asterisk). The leak was first detected by increased drain output with fluid of high creatinine concentration.
- **Fig. 61.24 Ultrasound image of a renal allograft** demonstrating hydronephrosis (H) secondary to extrinsic compression from a septated fluid collection — a lymphocele (L).